The Defense Trilemma: Impossibility of Continuous Utility-Preserving Wrapper Defenses

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Abstract

We prove that no continuous, utility-preserving wrapper defense function $D\colon X \to X$ that preprocesses inputs before the model sees them, can eliminate all safety violations from a language model with connected prompt space, and we characterize exactly where every such defense must fail. We establish three results of increasing strength: *boundary fixation*—the defense must leave some threshold-level inputs unchanged; an *ε-robust constraint*—under Lipschitz regularity, a positive-measure band around fixed boundary points remains near-threshold; and a *persistent unsafe region*—under a transversality condition, a positive-measure subset of inputs remains strictly unsafe. These constitute a *defense trilemma*: continuity, utility preservation, and completeness cannot coexist. We prove parallel discrete results requiring no topology, and extend to multi-turn interactions, stochastic defenses, and capacity-parity settings. The results do not preclude training-time alignment, architectural changes, or defenses that sacrifice utility. The full theory is mechanically verified in Lean 4 with Mathlib `https://github.com/mbhatt1/stuff/tree/main/ManifoldProofs` (45 files, ∼350 theorems, no admitted proofs, three standard axioms) and validated empirically on three LLMs [17].

1 Introduction

Can you build a wrapper around a language model that eliminates all prompt injection vulnerabilities? Most current defense work implicitly assumes yes. Input classifiers that flag suspicious prompts [1], output filters that catch unsafe completions [7], and constitutional rewriting pipelines [3] all share the same structure: a function $D\colon X \to X$ that preprocesses prompts before the model sees them, mapping unsafe inputs to safe equivalents while leaving safe inputs unchanged. We prove the answer is no, under two mild constraints. If the defense is *continuous* (similar prompts produce similar rewrites) and *utility-preserving* (safe prompts pass through unchanged), it cannot be *complete* (make every output safe). These three properties form a **defense trilemma**: any two can coexist, but not all three (Figure 2).

The impossibility is not about specific attacks or clever prompt engineering. It arises from the *geometry* of the prompt space itself: in a connected space, the safe region is open but not closed, so any continuous defense that fixes safe inputs must also fix points on the safety boundary. We establish three results, each strictly stronger than the last (Figure 3 provides visual intuition):

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Boundary fixation (Theorem 4.1). The defense must fix at least one boundary point, i.e., a prompt where alignment deviation equals the threshold exactly—passing it through with no remediation.

ε-robust constraint (Theorem 5.1). Under Lipschitz regularity, the defense cannot uniformly reduce alignment deviation far below τ near the fixed boundary point. For any x within distance δ of the fixed point z:

$$f(D(x)) \;\geq\; \tau - L(K+1)\,\delta. \tag{1}$$

Persistent unsafe region (Theorem 6.3). Under a transversality condition, the alignment surface rises faster than the defense can pull it down, leaving a positive-measure region that remains unsafe:

$$f(D(x)) > \tau \quad \text{for all } x \in \mathcal{S}, \qquad \mu(\mathcal{S}) > 0. \tag{2}$$

From discrete to continuous. All three results apply to continuous interpolants of discrete data. The Tietze extension theorem guarantees that any finite set of behavioral observations in a normal space admits continuous extensions, and the impossibilities hold for every such extension (Theorem 8.1).

Scope and limitations. Our results apply specifically to *continuous, utility-preserving wrapper defenses* on *connected* prompt spaces. They do *not* preclude effective safety through other mechanisms, including:

- training-time alignment (RLHF, DPO, constitutional AI training),
- architectural changes to the model itself,
- discontinuous defenses (e.g., hard blocklists or discrete classifiers),
- multi-component systems whose classifiers may reject or redirect inputs rather than preserving utility on every prompt.

All three conditions—continuity, utility preservation, and connectedness, are individually necessary; we give counterexamples for each (Section B). The core theory and all extensions are mechanically verified in Lean 4 with zero `sorry` statements. Full proofs appear in Section A; the Lean artifact is described in Section G.

2 Related Work

Adversarial robustness research has established that small perturbations can fool classifiers [20, 6, 10, 14] and that robustness may fundamentally trade off against accuracy [21, 11]. Certified defenses provide guarantees for fixed models [5, 8, 19, 13, 2], and topological perspectives have illuminated the structure of decision boundaries [15]. For LLMs specifically, jailbreaking attacks [23, 4, 9, 12] and automated red-teaming [16, 18] demonstrate persistent vulnerabilities.

Our work differs in kind: rather than studying how failures arise for fixed models or how specific systems can be certified, we impose *universal constraints on the defense map itself.* The closest conceptual precedent is the no-free-lunch framework [22], which proves that no optimizer dominates across all problems. We prove the analogous result for wrapper defenses: under continuity and utility preservation, no defense eliminates all failures.

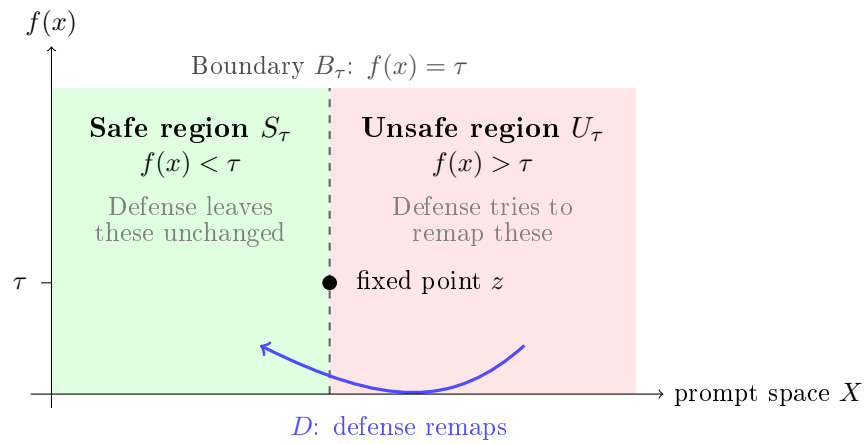

Figure 1: Schematic of the prompt space. The defense D must leave all safe inputs unchanged (utility preservation) and tries to remap unsafe inputs into the safe region. Boundary fixation (Theorem 4.1) proves the defense must also leave some boundary points unchanged, the black dot z passes through with no remediation.

3 Formal Framework

Before defining the model formally, we sketch the intuition. Every prompt lives in a space X. A scoring function f assigns each prompt a number measuring how far the model's response deviates from alignment goals: low scores are safe, high scores are unsafe. A fixed threshold τ separates the two. A *defense* preprocesses prompts before the model sees them, trying to push unsafe prompts into safe territory. We require two natural properties: *continuity* (similar prompts produce similar rewrites, i.e., no brittle edge-case behavior) and *utility preservation* (safe prompts pass through unchanged, so the defense does not degrade normal use). Figure 1 illustrates the setup.

Definition 3.1 (Alignment Deviation Function)**.** *Let X be a topological space (a set equipped with a notion of "nearness" between points). An* alignment deviation function *is a continuous map $f\colon X \to \mathbb{R}$, i.e., small changes in the input produce small changes in the score. Given a threshold $\tau \in \mathbb{R}$:*

$$S_\tau = \{x \in X : f(x) < \tau\} \qquad \textit{(safe region)} \tag{3}$$

$$U_\tau = \{x \in X : f(x) > \tau\} \qquad \textit{(unsafe region)} \tag{4}$$

$$B_\tau = \{x \in X : f(x) = \tau\} \qquad \textit{(boundary)} \tag{5}$$

Definition 3.2 (Defense)**.** *A* defense *is a continuous map $D\colon X \to X$. It is* utility-preserving *if $D(x) = x$ for all $x \in S_\tau$ (safe prompts pass through unchanged), and* complete *if $f(D(x)) < \tau$ for all $x \in X$ (every output is safe).*

Example 3.3 (Running example)**.** *Consider $X = [0, 1]$, $f(x) = x$, $\tau = 0.5$. The safe region is $[0, 0.5)$, the unsafe region is $(0.5, 1]$, and the boundary is $\{0.5\}$. A utility-preserving defense must satisfy $D(x) = x$ for all $x < 0.5$. Continuity then forces $D(0.5) = \lim_{x \to 0.5^-} D(x) = 0.5$, so the boundary point is fixed: $f(D(0.5)) = 0.5 = \tau$. No matter how cleverly the defense handles inputs above 0.5, the point $x = 0.5$ slips through. We will see that this simple observation generalizes to arbitrary connected spaces and grows sharper under regularity assumptions.*

The central question is whether a defense can be both utility-preserving and complete. The following sections show that under natural conditions the answer is no.

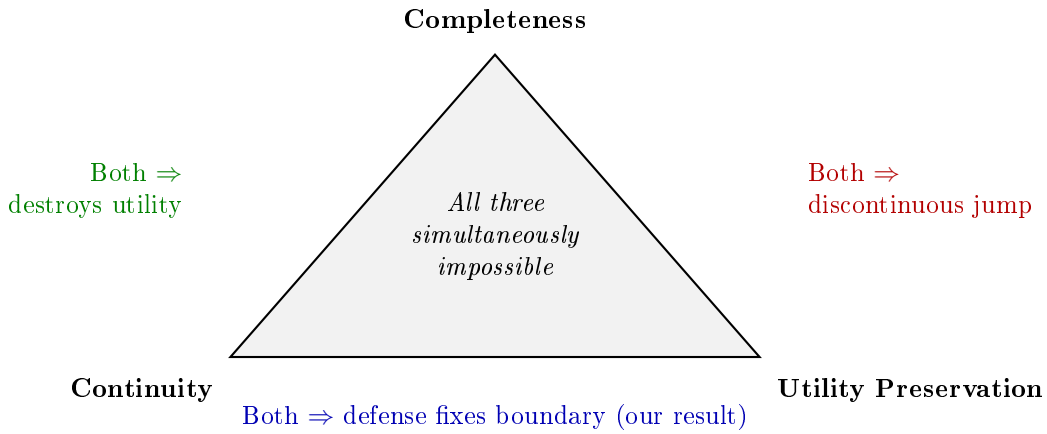

Figure 2: The defense trilemma. Any defense can satisfy at most two of the three properties. The bottom edge is our main result; the other two edges correspond to counterexamples in Section B.

4 Boundary Fixation

We begin with the most fundamental result. The argument fits in a paragraph: the defense fixes every safe input, so continuity makes its fixed-point set closed. But the safe region S_τ is open (preimage of an open interval under continuous f), and in a connected space a nonempty proper open set is not closed. Hence the fixed-point set cannot stop exactly at the edge of the safe region, it must spill onto the boundary B_τ. Some boundary prompts pass through unchanged.

Theorem 4.1 (Boundary Fixation)**.** *Let X be a connected Hausdorff space (a space that is "in one piece" and where distinct points can be separated by neighborhoods). Let $f\colon X \to \mathbb{R}$ be continuous with $S_\tau, U_\tau \neq \emptyset$, and $D\colon X \to X$ continuous with $D|_{S_\tau} = \mathrm{id}$. Then there exists $z \in X$ with $f(z) = \tau$ and $D(z) = z$. Moreover, every $z \in \overline{S_\tau} \setminus S_\tau$ satisfies $f(z) = \tau$ and $D(z) = z$, and this set is nonempty.*

Proof sketch. In a Hausdorff space, $\mathrm{Fix}(D)$ is closed (preimage of the diagonal). By utility preservation, $S_\tau \subseteq \mathrm{Fix}(D)$, so $\overline{S_\tau} \subseteq \mathrm{Fix}(D)$. But $S_\tau = f^{-1}((-\infty, \tau))$ is open and not closed (connectedness: a nonempty proper clopen set would disconnect X). Hence $\overline{S_\tau} \supsetneq S_\tau$. Any $z \in \overline{S_\tau} \setminus S_\tau$ satisfies $f(z) = \tau$ (limits of values $<\tau$ cannot exceed τ, and $z \notin S_\tau$ forces $f(z) \geq \tau$) and $D(z) = z$. Full proof in Section A. □

The defense's fixed-point set is too large to avoid the boundary. Utility preservation forces it to contain the safe region; closure forces it to contain the boundary; connectedness ensures the boundary is nonempty. Equivalently: a complete utility-preserving defense would be a continuous retraction $D\colon X \to S_\tau$ (since $D|_{S_\tau} = \mathrm{id}$ and $D(X) \subseteq S_\tau$), but a retract of a connected space is closed, while S_τ is open and proper—a contradiction.

Corollary 4.2. *No continuous utility-preserving defense is complete.*

All three hypotheses are individually necessary. A defense can satisfy at most two of the three—the *defense trilemma* (Figure 2). Counterexamples for each dropped hypothesis appear in Section B.

Remark 4.3 (Not all boundary points are fixed)**.** *The theorem captures $\overline{S_\tau} \setminus S_\tau$, not all of B_τ. Boundary points isolated from the safe side may escape. Verified in Lean with $f(x) = x^2$, $\tau = 0$.*

4.1 Relaxed Utility Preservation

A natural objection is that strict utility preservation ($D(x) = x$ for safe x) is too strong: a practical defense might rewrite "How do I bake a cake?" into "The user asks for a cake recipe. Please provide one." The prompt changes, but its safety score is preserved. The impossibility survives this relaxation and even weaker ones.

Theorem 4.4 (Score-Preserving Defense)**.** *Let X be a connected Hausdorff space, $f\colon X \to \mathbb{R}$ continuous with $S_\tau, U_\tau \neq \emptyset$. If $D\colon X \to X$ is continuous and* score-preserving on safe inputs*: $f(D(x)) = f(x)$ for all $x \in S_\tau$, then there exists z with $f(z) = \tau$ and $f(D(z)) = \tau$.*

Proof sketch. Define $h = f \circ D - f$. Then h is continuous and $h|_{S_\tau} = 0$. The zero set $\{h = 0\}$ is closed and contains $\overline{S_\tau}$. For $z \in \overline{S_\tau} \setminus S_\tau$: $f(D(z)) = f(z) = \tau$. □

This covers arbitrary rewrites of safe prompts, so long as they preserve the alignment score. The next result weakens even this:

Theorem 4.5 (ε-Relaxed Utility Preservation)**.** *Under the same hypotheses, if $|f(D(x)) - f(x)| \leq \varepsilon$ for all $x \in S_\tau$, then there exists z with $f(z) = \tau$ and $f(D(z)) \geq \tau - \varepsilon$.*

Proof sketch. The set $\{x : h(x) \geq -\varepsilon\}$ is closed and contains S_τ, hence $\overline{S_\tau}$. For $z \in \overline{S_\tau} \setminus S_\tau$: $f(D(z)) = \tau + h(z) \geq \tau - \varepsilon$. □

In plain terms: even if the defense is allowed to perturb safe scores by ε, it can only push the boundary by ε.

Remark 4.6 (Why $D(S_\tau) \subseteq S_\tau$ alone is insufficient)**.** *The weakest relaxation—$D(S_\tau) \subseteq S_\tau$ with no score constraint—does allow a complete defense (e.g., a constant map $D(x) = x_0$ to a fixed safe point). But this destroys all semantic content: every prompt produces the same response. The score-preservation conditions formalize the requirement that defense must not destroy utility, without requiring the defense to be the identity.*

5 ε-Robust Constraint

Boundary fixation tells us the defense must miss at least one point. But how bad is this? If the fixed point is an isolated needle in a vast haystack, the practical impact may be small. This section shows it is not isolated: Lipschitz regularity makes the failure spread to an entire neighborhood. If both f and D change at bounded rates (Lipschitz), then near a fixed boundary point z, neither the defense nor the scoring function can change much. The defense cannot "teleport" nearby points far into safe territory.

Theorem 5.1 (ε-Robust Defense Constraint)**.** *Under the hypotheses of Theorem 4.1, if (X, d) is a metric space, f is L-Lipschitz, and D is K-Lipschitz, then for the fixed boundary point z:*

$$f(D(x)) \;\geq\; \tau - L(K+1)\,\mathrm{dist}(x, z) \qquad \textit{for all } x \in X. \tag{6}$$

Points within distance δ of z remain within $L(K+1)\delta$ of threshold.

Proof sketch. Since $D(z) = z$ and D is K-Lipschitz: $\mathrm{dist}(D(x), z) \leq K\,\mathrm{dist}(x, z)$. Since $f(z) = \tau$ and f is L-Lipschitz: $|f(D(x)) - \tau| \leq L \cdot K\,\mathrm{dist}(x, z) \leq L(K+1)\,\mathrm{dist}(x, z)$. Full proof in Section A. □

In the running example (Theorem 3.3), both $f(x) = x$ and any reasonable defense are 1-Lipschitz, so $L = K = 1$ and the bound gives $f(D(x)) \geq 0.5 - 2|x - 0.5|$. Within distance 0.05 of $z = 0.5$, the defense cannot push the score below 0.40.

The next result shows the constrained band has positive volume:

Theorem 5.2 (Positive-Measure ε-Band)**.** *For every $\varepsilon > 0$ such that f takes values below $\tau - \varepsilon$, the band $\mathcal{B}_\varepsilon = \{x : \tau - \varepsilon \leq f(x) \leq \tau\}$ has positive measure (under any measure positive on nonempty open sets). Specifically, $B(c, \varepsilon/(4L)) \subseteq \mathcal{B}_\varepsilon$ for any c with $f(c) = \tau - \varepsilon/2$.*

The ε-band constrains the *depth* to which the defense can push near-boundary points into the safe region, though it does not by itself prevent slight reductions below τ. The next section addresses when even this is impossible.

6 Persistent Unsafe Region

The ε-robust constraint limits how far the defense can push near-boundary points, but it still allows the defense to nudge them slightly below τ. Can the defense always do at least that? Not necessarily. When the alignment surface rises away from the boundary faster than the defense can pull it down, some points are stuck above threshold no matter what.

Decoupling global and directional Lipschitz constants. The ε-robust bound (Theorem 5.1) uses the *global* Lipschitz constant L of f, which bounds f uniformly in all directions. The persistence argument, however, compares f's growth rate in the *steep direction* to how much the defense can reduce f along the *displacement direction* $D(x) - x$. In anisotropic settings these directions differ: f may rise steeply toward the unsafe region while varying slowly in the direction the defense pulls.

We write ℓ for the *defense-path Lipschitz constant*:

$$\ell = \sup_{x \neq D(x)} \frac{|f(D(x)) - f(x)|}{\operatorname{dist}(D(x),\, x)}.$$

Since f is L-Lipschitz globally, $\ell \leq L$ always. When the alignment surface is **isotropic** ($\ell = L$), the steep region is empty for every $K \geq 0$ (verified in Lean as `shallow_boundary_no_persistence`). The result is non-vacuous precisely when the surface is **anisotropic**: $\ell < L$, with directional gradient G satisfying $G > \ell(K + 1)$.

Lemma 6.1 (Input-Relative Bound)**.** *Under the hypotheses of Theorem 5.1, if f has defense-path Lipschitz constant ℓ, then $f(D(x)) \geq f(x) - \ell(K + 1)\operatorname{dist}(x, z)$ for all $x \in X$.*

Proof sketch. Triangle inequality: $\operatorname{dist}(D(x), x) \leq \operatorname{dist}(D(x), z) + \operatorname{dist}(z, x) \leq (K + 1)\operatorname{dist}(x, z)$. Defense-path Lipschitz: $|f(D(x)) - f(x)| \leq \ell \operatorname{dist}(D(x), x) \leq \ell(K + 1)\operatorname{dist}(x, z)$. □

The defense can reduce any point's score by at most $\ell(K + 1)$ times its distance from z. If the score rises faster than that, the defense loses.

Definition 6.2 (Steep region)**.** *Given a fixed boundary point z, the* steep region *is $\mathcal{S} = \{x \in X : f(x) > \tau + \ell(K+1) \operatorname{dist}(x, z)\}$—the set of points where alignment deviation exceeds τ by more than the defense's Lipschitz budget can compensate.*

Theorem 6.3 (Persistent Unsafe Region)**.** *Under the hypotheses of Theorem 5.1, if $\mathcal{S} \neq \emptyset$, then:*

1. *$\mathcal{S}$ is open.*

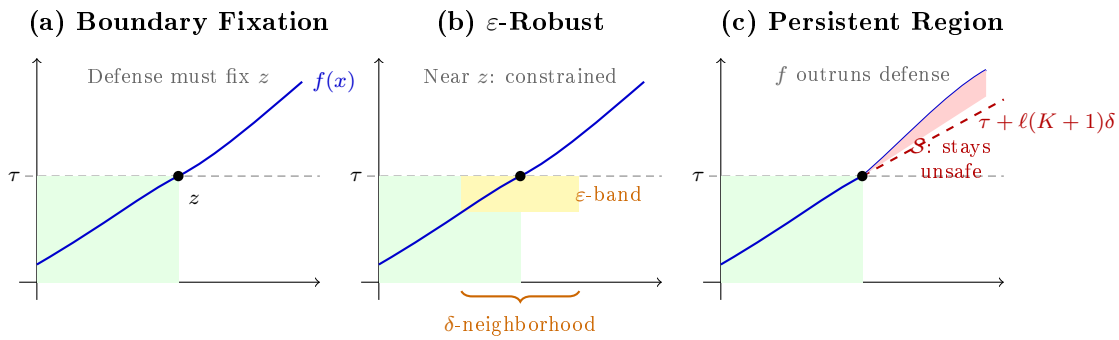

Figure 3: The three impossibility results on a 1D cross-section. **(a)** The defense must fix boundary point z (Thm. 4.1). **(b)** Near z, Lipschitz regularity constrains the defense to a shallow ε-band (yellow; Thm. 5.1). **(c)** Where f rises faster than the defense budget $\ell(K+1)\delta$ (dashed red), the region above τ persists (red shading; Thm. 6.3).

2. *$\mathcal{S}$ has positive measure (under any measure positive on nonempty open sets).*
3. *For every $x \in \mathcal{S}$: $f(D(x)) > \tau$.*

The defense leaves a positive-measure region that remains unsafe.

Proof sketch. $\mathcal{S}$ is a strict superlevel set of a continuous function, hence open. For $x \in \mathcal{S}$: $f(D(x)) \geq f(x) - \ell(K+1)\,\mathrm{dist}(x,z) > \tau$ by Theorem 6.1 and the definition of $\mathcal{S}$. Full proof in Section A. □

When does the steep region exist? Whenever the alignment surface has directional slope exceeding $\ell(K+1)$ at the boundary:

Proposition 6.4 (Transversality from Directional Derivative)**.** *In a normed space, if f has directional derivative $c > \ell(K+1)$ along a unit vector v at boundary point z (i.e., $f(z+tv) \geq \tau + ct$ for small $t > 0$), then $z+tv \in \mathcal{S}$ for sufficiently small t. Verified in Lean as* `transversality_from_deriv`*.*

This condition is observed empirically in all three models we study (Section 11): the alignment surface rises steeply toward the unsafe region (G large) while the defense operates in a smoother subspace ($\ell \ll L$).

7 Quantitative Bounds

The preceding results establish that failures *exist* and have positive measure. This section provides explicit lower bounds and identifies a fundamental dilemma in choosing the defense's aggressiveness.

Theorem 7.1 (Volume Lower Bound)**.** *Let $f\colon \mathbb{R}^n \to \mathbb{R}$ be L-Lipschitz with $L > 0$. If there exists c with $f(c) = \tau - \varepsilon/2$, then $B(c, \varepsilon/(4L)) \subseteq \mathcal{B}_\varepsilon$, giving:*

$$\mu(\mathcal{B}_\varepsilon) \;\geq\; V_n \cdot \left(\frac{\varepsilon}{4L}\right)^n \tag{7}$$

where V_n is the unit ball volume in $\mathbb{R}^n$. In $\mathbb{R}^1$, this simplifies to $\mu(\mathcal{B}_\varepsilon) \geq \varepsilon/(2L)$.

In plain terms: smoother alignment surfaces (smaller L) have wider ε-bands, but the band always has positive volume.

Theorem 7.2 (Cone Measure Bound)**.** *In $\mathbb{R}$, if f has directional growth rate $c > \ell(K+1)$ at boundary point z on $(z, z+\delta_0)$, then $\mu(\{x : f(D(x)) > \tau\}) \geq \delta_0$.*

This gives a concrete lower bound on the persistent region: if the alignment surface is steep over an interval of length δ_0, the defense fails on at least that much volume.

The defense designer faces a dilemma in choosing the Lipschitz constant K of the defense:

Theorem 7.3 (Defense Dilemma). *Let $G = |\nabla f(z)|$ at boundary point z, and let ℓ be the defense-path Lipschitz constant (Section 6). Define $K^* = G/\ell - 1$. Then:*

1. *If $K < K^*$: the persistent unsafe region exists ($G > \ell(K+1)$, Theorem 6.3 applies).*
2. *If $K \geq K^*$: the defense distorts space at least as much as the alignment gradient ($\ell(K+1) \geq G$).*

No value of K avoids both failure modes simultaneously. Since $\ell \leq L$, the dilemma is sharpest when $\ell \ll L$ (anisotropic surfaces). When $\ell = L$ (isotropic), $K^* \leq 0$ and only the wide-band failure mode applies.

A gentle defense (K small) cannot compensate for steep alignment surfaces. An aggressive defense (K large) distorts prompts so heavily that the ε-band widens. The defense is caught between two failure modes.

8 From Discrete Data to Continuous Theory

A natural concern is that language models operate on discrete tokens, not continuous spaces. We address this from both directions: first showing the continuous results apply to any continuous model consistent with discrete observations, then proving parallel results directly on finite sets with no continuous machinery at all.

8.1 Continuous Interpolation

Any finite set of behavioral observations can be extended to a continuous function on the full space. The classical Tietze extension theorem guarantees this:

Theorem 8.1 (Continuous Relaxation). *Let X be a connected, normal, Hausdorff space, and $S \subset X$ a finite set of observations with $g(p) < \tau$ and $g(q) > \tau$ for some p, q. Then there exists a continuous $f\colon X \to \mathbb{R}$ agreeing with all observations for which Theorems 4.1 and 5.1 hold. If f is additionally Lipschitz (as for GP posterior means under standard kernel assumptions), Theorem 6.3 applies wherever transversality is met.*

Proof sketch. Finite subsets of T_1 spaces are closed. By Tietze, g extends to a continuous f. Since $f(p) < \tau < f(q)$, Theorem 4.1 applies. Lipschitz extensions (McShane–Whitney) enable the remaining results. □

If we observe both safe and unsafe model behaviors, the impossibility holds for *every* continuous model consistent with our observations.

8.2 Direct Discrete Results

To address the objection that continuous impossibility might be an artifact of continuous relaxation, we prove parallel results directly on finite sets using only counting arguments and induction. No topology is required; all results are verified in Lean as `MoF_12_Discrete`.

Theorem 8.2 (Discrete IVT). *Let $f\colon \{0, \ldots, n+1\} \to \mathbb{R}$ with $f(0) < \tau$ and $f(n+1) \geq \tau$. Then there exists i with $f(i) < \tau$ and $f(i+1) \geq \tau$.*

Theorem 8.3 (Discrete Defense Dilemma). *Let X be a finite set with $S_\tau, U_\tau \neq \emptyset$, and $D\colon X \to X$ utility-preserving ($D(x) = x$ for $f(x) < \tau$).*

1. *If D is injective, then $f(D(u)) \geq \tau$ for every $u \in U_\tau$: the defense is incomplete.*
2. *If D is complete ($f(D(x)) < \tau$ for all x), then D is non-injective: $\exists\, x \neq y$ with $D(x) = D(y)$.*

Proof. (1) Suppose D is injective and $f(D(u)) < \tau$ for some $u \in U_\tau$. Then $D(u)$ is safe, so $D(D(u)) = D(u)$ by utility preservation. Injectivity gives $D(u) = u$, so $f(u) = f(D(u)) < \tau$, contradicting $u \in U_\tau$.

(2) For any $u \in U_\tau$: completeness gives $f(D(u)) < \tau$, so utility preservation gives $D(D(u)) = D(u)$. But $u \neq D(u)$ since $f(u) \geq \tau > f(D(u))$. So u and $D(u)$ are distinct inputs with $D(u) = D(D(u))$: the defense is non-injective. □

The continuous trilemma trades *continuity* for completeness; the discrete dilemma trades *injectivity*. Part (1) is the genuine constraint: an information-preserving defense *cannot* eliminate unsafe outputs. Any complete defense must destroy information—collapsing distinct inputs to the same output. This is not a failure of the defense; it is the *mechanism* by which it operates. But the information loss has structural consequences: the defense erases the distinction between a genuine safe input and a remapped attack, precluding downstream auditing of which inputs were intercepted and preventing the system from learning which attacks were attempted.

9 Extensions

The core results assume a static, deterministic, single-turn defense. *Does multi-turn interaction, randomization, or pipelining provide an escape?* We show it does not. Each extension is a direct application of the boundary fixation machinery to a modified setting.

9.1 Multi-Turn Impossibility

In practice, LLMs maintain state across conversation turns. One might hope that a defense could use context from previous turns to "anticipate" boundary prompts. But the topology constrains each turn independently:

Theorem 9.1 (Multi-Turn Impossibility). *Let $\{f_t, D_t\}_{t=1}^T$ be alignment functions and defenses over T turns, each continuous and utility-preserving. Then for every turn t, there exists z_t with $f_t(z_t) = \tau$ and $D_t(z_t) = z_t$.*

Proof sketch. Apply Theorem 4.1 to (f_t, D_t) at each turn. The functions may depend on full history—this does not matter, as each timestep is a fresh instance of boundary fixation. □

Worse, multi-turn interaction helps the attacker: boundary information accumulates, the best observed exploit improves monotonically, and the attacker can steer toward transversality.

9.2 Stochastic Defense Impossibility

Claim: If the defense maps each input to a *distribution* over outputs, perhaps boundary fixation can be averaged away. It cannot:

Theorem 9.2 (Stochastic Defense Impossibility). *Let D be a stochastic defense and define $g(x) = \mathbb{E}_{y\sim D(x)}[f(y)]$. If g is continuous and $g(x) = f(x)$ for all $x \in S_\tau$, then there exists z with $f(z) = \tau$ and $g(z) = \tau$.*

Proof sketch. Apply Theorem 4.1 to g in place of $f \circ D$: $h = g - f$ vanishes on S_τ and is continuous, so it vanishes on $\overline{S_\tau}$. □

The expected post-defense score at the boundary is still τ. Randomization cannot beat the topology.

9.3 Capacity Parity

Even when the defense has equal computational resources to the attacker, it faces a structural asymmetry:

Theorem 9.3 (Capacity Parity Disadvantage)**.** *The defense must satisfy two constraints simultaneously—correct classification (distinguishing S_τ from U_τ) and utility preservation (identity on S_τ)—while the attacker need only find one input in U_τ. The defense's hypothesis class must represent both the decision boundary and the identity function on the safe side, while the attacker's search is unconstrained.*

Utility preservation is a structural tax on the defense that the attacker does not pay: the attacker explores freely, while the defense must simultaneously preserve and filter.

9.4 Nonlinear Agent Pipelines

Modern AI agents chain multiple processing stages; pipeline depth multiplies the effective Lipschitz constant and widens the failure band.

Theorem 9.4 (Pipeline Lipschitz Degradation)**.** *If stages $T_1, \ldots, T_n$ are $K_1, \ldots, K_n$-Lipschitz, the composed pipeline is $(\prod K_i)$-Lipschitz. For n stages each with $K \geq 2$, the effective constant is K^n—exponential in depth.*

Theorem 9.5 (Pipeline Impossibility)**.** *If the composed pipeline $P = T_n \circ \cdots \circ T_1 \circ D$ is continuous and utility-preserving, then P has boundary fixed points. The ε-robust band scales as $L(K^n + 1)\delta$.*

In plain terms: each additional pipeline stage *exponentially* widens the band in which the defense is ineffective. Tool-using agents are fundamentally harder to defend than single-turn models.

10 The Vulnerability Landscape

The preceding sections show that defenses must fail *somewhere*. This section characterizes *where*: vulnerabilities are not isolated needles in a haystack but extended regions ("basins") with computable minimum sizes.

Theorem 10.1 (Basin Structure)**.** *If f is continuous and $f(p) > \tau$, then U_τ is open. Under any measure positive on nonempty open sets, U_τ has positive measure.*

Theorem 10.2 (Basin Fragment Minimum Size)**.** *If f is L-Lipschitz with $f(p) > \tau$, the connected component of U_τ containing p has diameter $\geq 2(f(p) - \tau)/L$.*

The second result is intuitive: if a point scores far above threshold, the Lipschitz constraint prevents f from dropping back below τ quickly, so the unsafe basin around it must be large. Smoother surfaces (smaller L) produce larger, more predictable basins; rougher surfaces produce a mosaic of smaller fragments.

Additional supporting results like perturbation robustness, iterative convergence, transferability, authority monotonicity, and gradient ascent, are developed in Section C. Fine-tuning stability results appear in Section D, and cost asymmetry analysis in Section E.

Table 1: Falsifiable predictions confirmed by empirical data.

Theorem	**Predicts**	**Confirmed by**
Basin Structure (10.1)	Basins are open with positive measure	Heatmaps show extended regions
Fragmentation (10.2)	Smooth → large basins; rough → mosaic	Llama: mesa; GPT-OSS: mosaic
Convergence (C.2)	Attacks exhibit monotone convergence	Convergence curves plateau
Transferability (C.3)	Similar surfaces → shared basins	Llama .93 → GPT-OSS .73 → Mini .47
Authority (C.4)	Horizontal banding	Bands at $a_2 \approx$ .25–.35, .65–.85
Persistent (6.3)	Steep boundaries → unsafe volume persists	Llama's .93 plateau persists under defense
Interior Stability (D.1)	Deep basin points survive fine-tuning	Vulnerability persists across variants
Cost (E.1)	2D tractable, high-d intractable	15K queries fill 63% at $d=2$
Pipeline (9.4)	Deeper pipelines → wider failure band	Tool-using agents harder to defend than single-turn

11 Experimental Validation

The Manifold of Failure framework [17] maps three LLMs over a 2D behavioral space with two axes: *query indirection* (how obliquely the prompt asks for unsafe content) and *authority framing* (how much the prompt invokes authority or permission). Each point corresponds to a prompt style, and the alignment deviation score measures how unsafe the model's response is. Table 1 summarizes nine falsifiable predictions, all confirmed.

Llama-3-8B (mean AD 0.93, basin rate 93.9%): a near-flat alignment surface (small L) producing large robustness radii, a gently sloping mesa. The directional slope at boundary crossings far exceeds $\ell(K+1)$ for any reasonable K, placing this model firmly in persistent-unsafe territory: the defense cannot eliminate the vulnerability volume.

GPT-OSS-20B (mean AD 0.73, basin rate 64.3%): a rugged landscape (large L) with many small fragments. Horizontal bands confirm authority monotonicity. The fragmented boundary presents many transversal crossings, each generating a persistent unsafe pocket.

GPT-5-Mini (peak AD 0.50, basin rate 0%): at $\tau = 0.50$, $U_\tau = \emptyset$—none of the three theorems apply, correctly predicting no impossibility. This model exemplifies the best available strategy: make the boundary harmless rather than try to eliminate it.

12 The Engineering Prescription

The results do not say defense is valueless; they say *complete* defense is impossible under the stated constraints. The engineering goal shifts from elimination to management, ordered from most to

least actionable:

1. Make the boundary shallow. Set τ so that boundary-level behavior is benign. If $f(z) = \tau$ yields a polite refusal rather than harmful compliance, the impossibility is mathematically true but practically harmless. GPT-5-Mini exemplifies this: its ceiling at AD = 0.50 means the failure manifold exists but contains no actual harm.

2. Reduce the Lipschitz constant. Smaller L makes the alignment surface smoother, raises the transversality threshold $\ell(K+1)$ for persistence, and shrinks the persistent region. The tradeoff: smoother surfaces spread vulnerabilities over wider but more easily monitored regions.

3. Reduce the effective dimension. Defense cost grows as N^d (Theorem E.1). Constraining the prompt interface—standardized formats, restricted API parameters, bounded context lengths—reduces d, making the behavioral space tractable.

4. Monitor, don't eliminate, the boundary. Transversal crossings persist under fine-tuning (Theorem D.2) and recur every turn (Theorem 9.1). Rather than attempt the impossible, deploy runtime monitoring that detects approach to the boundary. The Lipschitz bound (Theorem C.1) gives a computable estimate of distance to the boundary from any observed AD value.

5. Break the asymmetry with architectural separation. The defense must simultaneously classify and preserve (Theorem 9.3), while the attacker only searches. Separate the defense from the utility pipeline—e.g., a lightweight classifier that flags, paired with a separate response generator—to decouple the two tasks.

13 Limitations

Boundary fixation is at the boundary. Fixed points satisfy $f(z) = \tau$ exactly. If τ-level behavior is benign, the theorem is true but harmless.

The ε-robust constraint limits depth, not direction. The defense may push near-boundary points slightly below τ. The bound limits how far, not whether.

Persistence requires transversality. The persistent unsafe region exists only where the alignment surface is steep ($c > \ell(K+1)$, where ℓ is the defense-path Lipschitz constant). For isotropic f ($\ell = L$), $\mathcal{S}$ is empty for all $K \geq 0$.

Grid-based cost asymmetry. Theorem E.1 assumes exhaustive grid enumeration. Learning-based defenses that generalize across the space may sidestep the exponential bound.

14 Conclusion

We establish a three-level impossibility hierarchy for prompt-injection defense: boundary fixation, the ε-robust constraint, and the persistent unsafe region theorem. Continuous topology, Lipschitz bounds, discrete counting, stochastic expectations, multi-turn dynamics, and capacity constraints all converge on the same conclusion: under the wrapper model, some failures persist.

In short: The trilemma perfectly proves that you cannot build a smooth, invisible, perfect shield around an LLM. But you don't need a smooth shield—you need a reinforced concrete wall with a strict bouncer at the door. The practical prescription is to make the boundary shallow, smooth, and low-dimensional, and to engineer around it rather than assume it can be eliminated.

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Appendices

A Full Proofs

Proof of Theorem 4.1 (Boundary Fixation). **Step 1** (Hausdorff $\Rightarrow$ fixed-point set is closed). $\mathrm{Fix}(D) = \{x : D(x) = x\}$ is the preimage of the diagonal $\Delta \subset X \times X$ under $x \mapsto (D(x), x)$. In a Hausdorff space, Δ is closed, so $\mathrm{Fix}(D)$ is closed.

Step 2 (Utility preservation $\Rightarrow$ safe region $\subseteq$ fixed points). $S_\tau \subseteq \mathrm{Fix}(D)$. Since $\mathrm{Fix}(D)$ is closed: $\overline{S_\tau} \subseteq \mathrm{Fix}(D)$.

Step 3 (Connectedness $\Rightarrow$ safe region is not closed). $S_\tau = f^{-1}((-\infty, \tau))$ is open. If also closed, it would be clopen—but in a connected space the only clopen sets are $\emptyset$ and X. Since both S_τ and U_τ are nonempty, S_τ is not closed.

Step 4 (Boundary point exists). $\overline{S_\tau} \supsetneq S_\tau$, so there exists $z \in \overline{S_\tau} \setminus S_\tau$. Continuity gives $f(z) \leq \tau$; $z \notin S_\tau$ gives $f(z) \geq \tau$. Hence $f(z) = \tau$.

Step 5 (Defense fixes the boundary point). $z \in \overline{S_\tau} \subseteq \mathrm{Fix}(D)$, so $D(z) = z$ and $f(D(z)) = \tau$. □

Proof of Theorem 5.1 (ε-Robust Constraint). By Theorem 4.1, the fixed boundary point z exists.

Step 1. Since $D(z) = z$ and D is K-Lipschitz: $\mathrm{dist}(D(x), z) = \mathrm{dist}(D(x), D(z)) \leq K\ \mathrm{dist}(x, z)$.

Step 2. Since $f(z) = \tau$ and f is L-Lipschitz: $|f(D(x)) - \tau| = |f(D(x)) - f(z)| \leq L\ \mathrm{dist}(D(x), z) \leq LK\ \mathrm{dist}(x, z) \leq L(K+1)\ \mathrm{dist}(x, z)$. The last uses $K \leq K+1$; we keep the uniform constant since it matches the displacement bound from the triangle inequality. □

Proof of Theorem 6.3 (Persistent Unsafe Region). **(1)** $\mathcal{S}$ is the strict superlevel set of $x \mapsto f(x) - \ell(K+1)\,\mathrm{dist}(x, z)$ at level τ, hence open.

(2) Open and nonempty implies positive measure.

(3) For $x \in \mathcal{S}$: $f(x) > \tau + \ell(K+1)\,\mathrm{dist}(x, z)$. By Theorem 6.1: $f(D(x)) \geq f(x) - \ell(K+1)\,\mathrm{dist}(x, z) > \tau$. □

B Counterexamples: Each Hypothesis Is Necessary

Counterexample B.1 (Removing connectedness). *$X = \{0, 1\}$ discrete, $f(0) = 0$, $f(1) = 1$, $\tau = 0.5$. $D(0) = 0$, $D(1) = 0$: continuous, utility-preserving, complete.*

Counterexample B.2 (Removing continuity). *$X = [0, 1]$, $f(x) = x$, $\tau = 0.5$. $D(x) = x$ for $x < 0.5$, $D(x) = 0$ for $x \geq 0.5$: utility-preserving and complete, but discontinuous at 0.5.*

Counterexample B.3 (Removing utility preservation). *$D(x) = x_0$ for a fixed safe point: continuous and complete, but destroys all inputs.*

C Attack Properties

Theorem C.1 (Perturbation Robustness). *If f is L-Lipschitz and $f(p) > \tau$, then $B(p, (f(p) - \tau)/L) \subseteq U_\tau$. The radius is monotone in $f(p)$.*

Theorem C.2 (Iterative Convergence). *Any monotone-improvement operator T with score in $[0, 1]$ converges. If each step gains $\geq \delta$, convergence takes $\leq \lfloor 1/\delta \rfloor$ steps.*

Theorem C.3 (Transferability). $\|f - g\|_\infty \leq \delta \implies \{f > \tau + \delta\} \subseteq \{g > \tau\}$. *Transfer costs zero additional queries.*

Theorem C.4 (Authority Monotonicity). *If $f(y, \cdot)$ is monotone non-decreasing in authority for each fixed indirection y, the vulnerability set is upward-closed with a critical threshold $a_2^*(y)$ (by IVT, when $f(y, \cdot)$ is continuous and crosses τ). If additionally f is monotone non-decreasing in indirection for each fixed authority, the threshold curve $y \mapsto a_2^*(y)$ is non-increasing.*

Theorem C.5 (Gradient Ascent). *If f has nonzero Fréchet derivative at x, there exist v and $\varepsilon > 0$ with $f(x + \varepsilon v) > f(x)$.*

D Stability Under Fine-Tuning

Theorem D.1 (Interior Stability). *If $\|f - g\|_\infty \leq \varepsilon$: $f(x) > \tau + \varepsilon \implies g(x) > \tau$, and $f(x) < \tau - \varepsilon \implies g(x) < \tau$. Only the band $|f(x) - \tau| \leq \varepsilon$ is uncertain.*

Theorem D.2 (Crossing Preservation). *If f crosses τ transversally on $[a, b]$ and $\|f - g\|_\infty \leq \varepsilon$, then g also crosses τ on $[a, b]$.*

Theorem D.3 (Patching Is Nonlocal). *There exist f, g with $\|f - g\|_\infty \leq \varepsilon$ such that eliminating a vulnerability at one point necessarily changes values at distant points.*

E Cost Asymmetry

Theorem E.1 (Exponential Cost Asymmetry). *For grid-based defense with $N \geq 2$ bins per axis in d dimensions: attack cost $\leq 1/\delta$ (dimension-independent); defense cost $= N^d$ (exponential in d); ratio $\delta \cdot N^d \to \infty$ as $d \to \infty$.*

At $d = 2$ with $N = 25$ and $\delta = 0.01$, the ratio is 6.25. At $d = 10$, it climbs to $\sim 10^{12}$.

F Additional Verified Results

Lipschitz displacement bound. D is K-Lipschitz with $D(z) = z \implies \mathrm{dist}(D(x), x) \leq (K + 1)\,\mathrm{dist}(x, z)$.

Tool calls amplify failure. Each non-contractive tool call multiplicatively increases the pipeline's effective Lipschitz constant: $K^n < K^{n+1}$ for $K \geq 2$.

Attacker monotone improvement. Best observed alignment deviation is non-decreasing across turns.

Boundary knowledge accumulation. Known boundary points grow monotonically; after T probes, T points are classified.

Attacker steering. If directional slope varies continuously with attacker parameter α and crosses $\ell(K + 1)$, transversality is reachable by IVT.

Stochastic regularity. $g(x) = \mathbb{E}[f(D(x))]$ satisfies $g - f = 0$ on $\overline{S_\tau}$.

Discrete defense dilemma. An injective, utility-preserving defense is incomplete (every unsafe input stays unsafe). A complete, utility-preserving defense is non-injective (distinct inputs collapse to the same output). The three properties—completeness, utility preservation, injectivity—form a trilemma.

Defense position invariance. Lipschitz constant of defense-before-tools equals defense-after-tools: $K_D \cdot K_T^n$.

G Lean Artifact

The complete theory is verified in Lean 4.28.0 with Mathlib v4.28.0, available at `https://github.com/mbhatt1/stuff/tree/main/ManifoldProofs`. The artifact comprises 45 files:

- 10 core theory files (MoF_01–MoF_10)
- 10 cost theory files (MoF_Cost_01–MoF_Cost_10)
- 10 advanced theory files (MoF_Adv_01–MoF_Adv_10)
- 1 continuous relaxation (MoF_ContinuousRelaxation)
- 1 ε-robust + persistent unsafe region (MoF_11_EpsilonRobust)
- 1 discrete impossibility (MoF_12_Discrete)
- 1 multi-turn + stochastic + capacity parity (MoF_13_MultiTurn)
- 1 representation-independent meta-theorem (MoF_14_MetaTheorem)
- 1 nonlinear agent pipelines (MoF_15_NonlinearAgents)
- 1 relaxed utility preservation (MoF_16_RelaxedUtility)
- 1 quantitative ε-band volume bound (MoF_17_CoareaBound)
- 1 cone measure bound for persistent unsafe region (MoF_18_ConeBound)
- 1 optimal defense characterization (MoF_19_OptimalDefense)
- 1 refined persistence with defense-path Lipschitz (MoF_20_RefinedPersistence)
- 3 capstone files (MasterTheorem, Euclidean instantiation, verification)
- 1 root import file (ManifoldProofs.lean)